# Module 8

Quick Reference and Course Summary

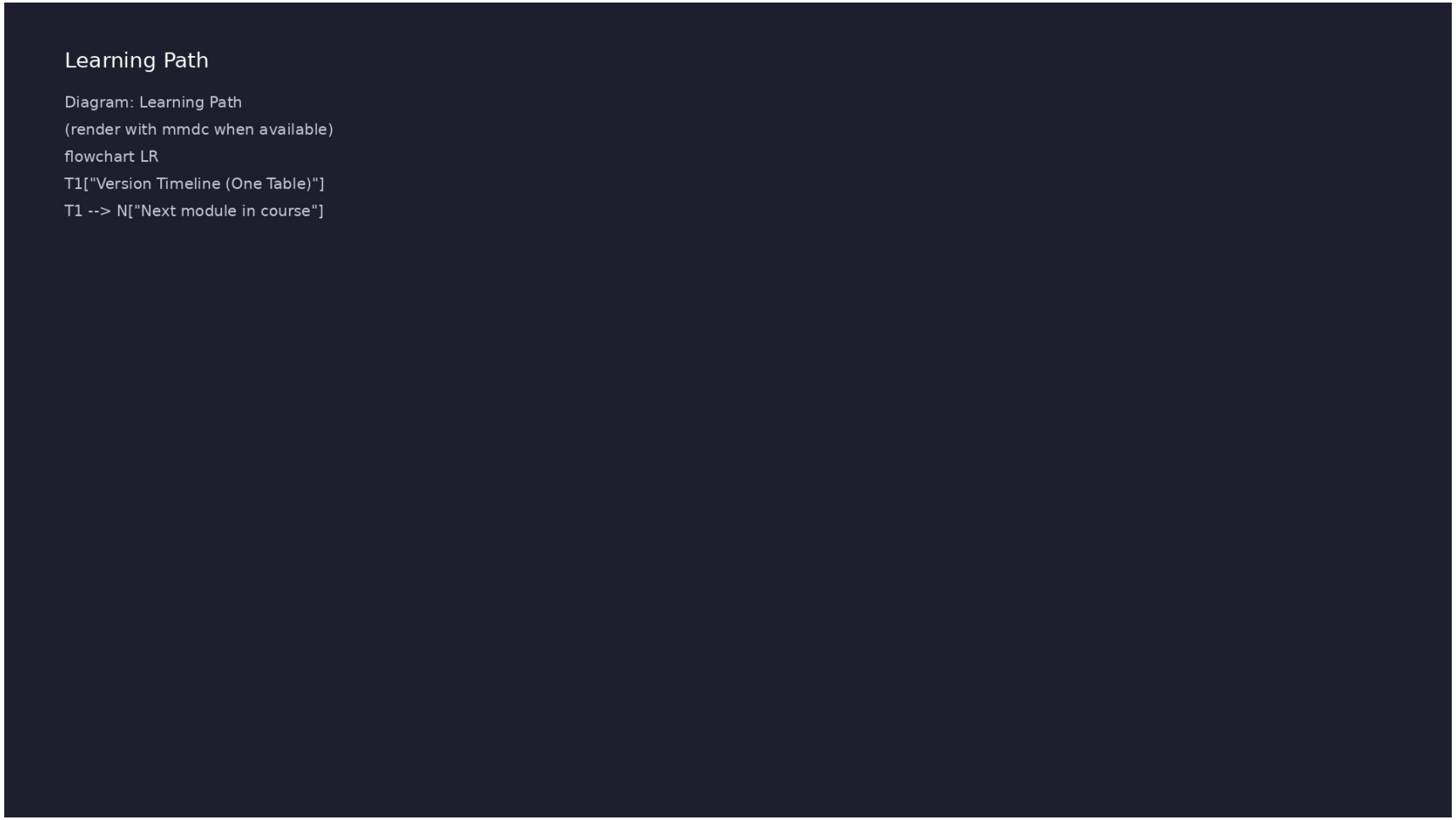
# Learning objectives

- Provide a single quick reference and course summary for Verilog and SystemVerilog across IEEE versi...

# Prerequisites

- See module README

# Learning path




Provide a single quick reference and course summary for Verilog and SystemVerilog across IEEE versions (1364-1995 through 1800-2017).

# Overview

- This module is a **reference and summary** for the version-centric Verilog/SystemVerilog course (Mo…

# Design architecture

# RTL / block architecture


Module 8: DUT structure and key interfaces (see common_dut/rtl/).

# Verification environment architecture


Module 8: UVM layers, agents, and checkers for this phase.

# 1. Course knowledge architecture (not a new DUT)

- Module 8 is a meta-layer: maps Modules 1–7 to standards, constructs, and learning path
- No new RTL — architecture here is how reference material is organized, not a chip block
- Sections: version timeline → construct-by-standard → design subset → migration → tool support
- Runnable examples (`module8/examples/`): each prints one slice of the quick reference for terminal…

# 2. How design subsets relate across the course

- Verilog-only path: Modules 1–3 (1364-1995 → 2001 → 2005) — `wire`/`reg`, `always @*`
- SystemVerilog design path: Modules 4–6 — `logic`, `always_comb`/`always_ff`, interfaces, packages
- Integration path: Module 7 side-by-side pairs; Module 8 tables tie constructs to modules
- Choosing a subset: project README should state revision + allowed construct list from Module 8 tabl…

# 3. Reference-to-RTL traceability

- construct_lookup example: answers “which module introduced X?” — links study back to Modules 1–6 DU…
- course_map example: module titles map to `docs/MODULEn.md` and `moduleN/dut/` where applicable
- synthesizable_subset example: do/avoid rules apply to all prior DUT architectures

# Verification & testing methods

# Testing and closure flow


Module 8: how tests, checks, and metrics close verification goals.

# 1. Verification strategy without new testbenches

- Module 8 does not add tests — it documents how to
  test when applying course knowledge
- Regression rule: after picking a standard from
  Module 8 tables, run the owning module's
  `./scripts/…
- Migration rule: use Module 7 equivalence tests
  whenever syntax changes under stable behavior
- Subset rule: CI/lint flags should match the
  documented Verilog-only vs SV-design subset

# 2. Lookup-driven test planning

- Before using a construct, run construct_lookup / Section 2 table — note minimum module and revision
- pitfalls example: common mistakes (wrong standard, mixed subsets) become test-plan checklist items
- version_selection example: document simulator/synthesis flags next to chosen revision in project RE…

# 3. Closure and ongoing maintenance

- learning_path example: ordered study 1→6, then 7; use 8 during real projects as guardrails
- When a new IEEE revision ships: add Module N, update Module 7 pairs and Module 8 tables — re-run al…
- Golden rule: Module 8 guides *what* to run; Modules 1–7 provide the actual tests and DUTs

# Syllabus topics

# 1. Version Timeline (One Table)

- See docs/MODULE8.md

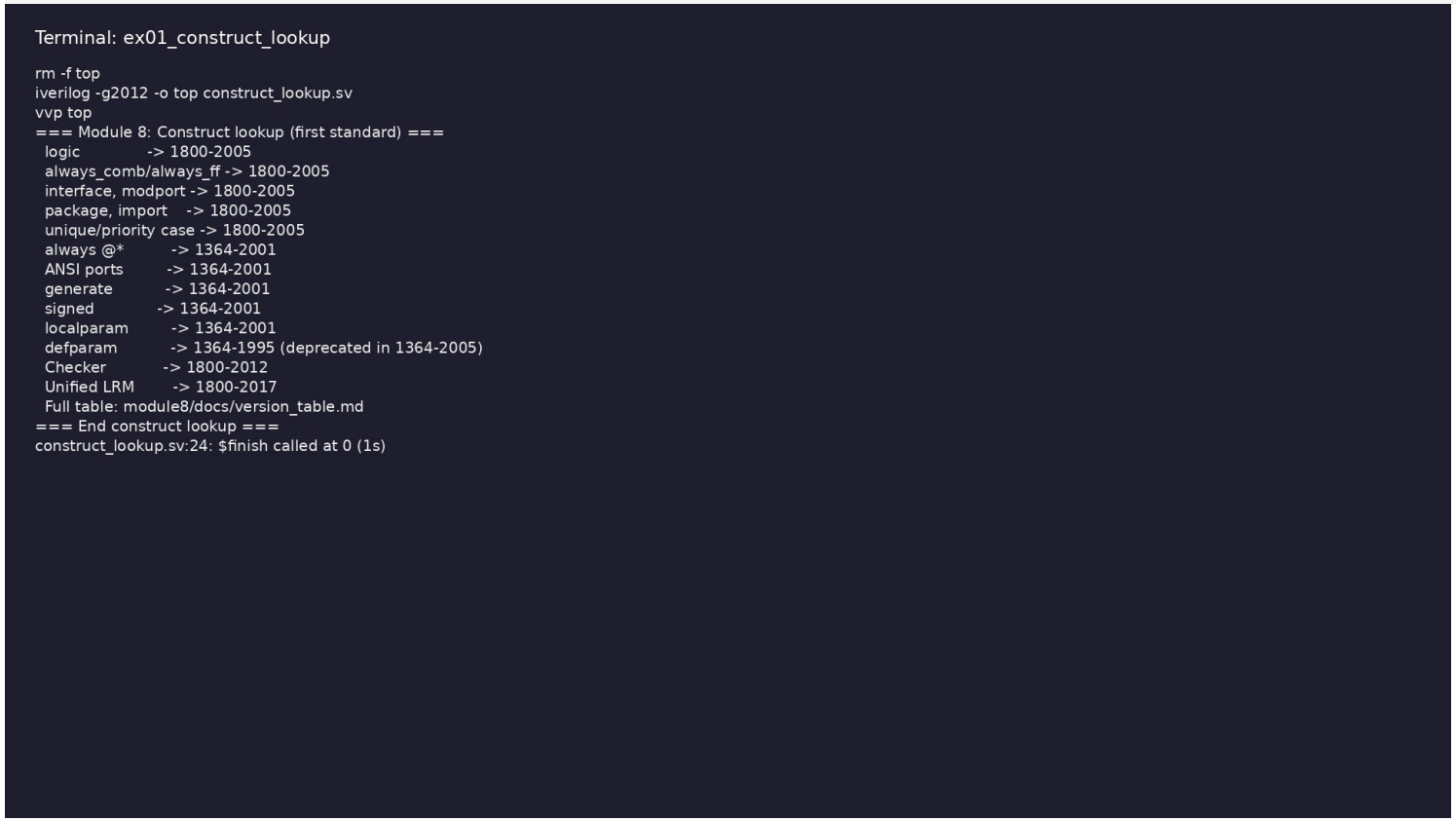
# Hands-on examples

# Module 8 self-check

```
$ ./scripts/module8.sh --check
```

# Demo: Construct Lookup

```
$ cd module8/examples/construct_lookup && make clean && make run
```



# Demo: Course Map

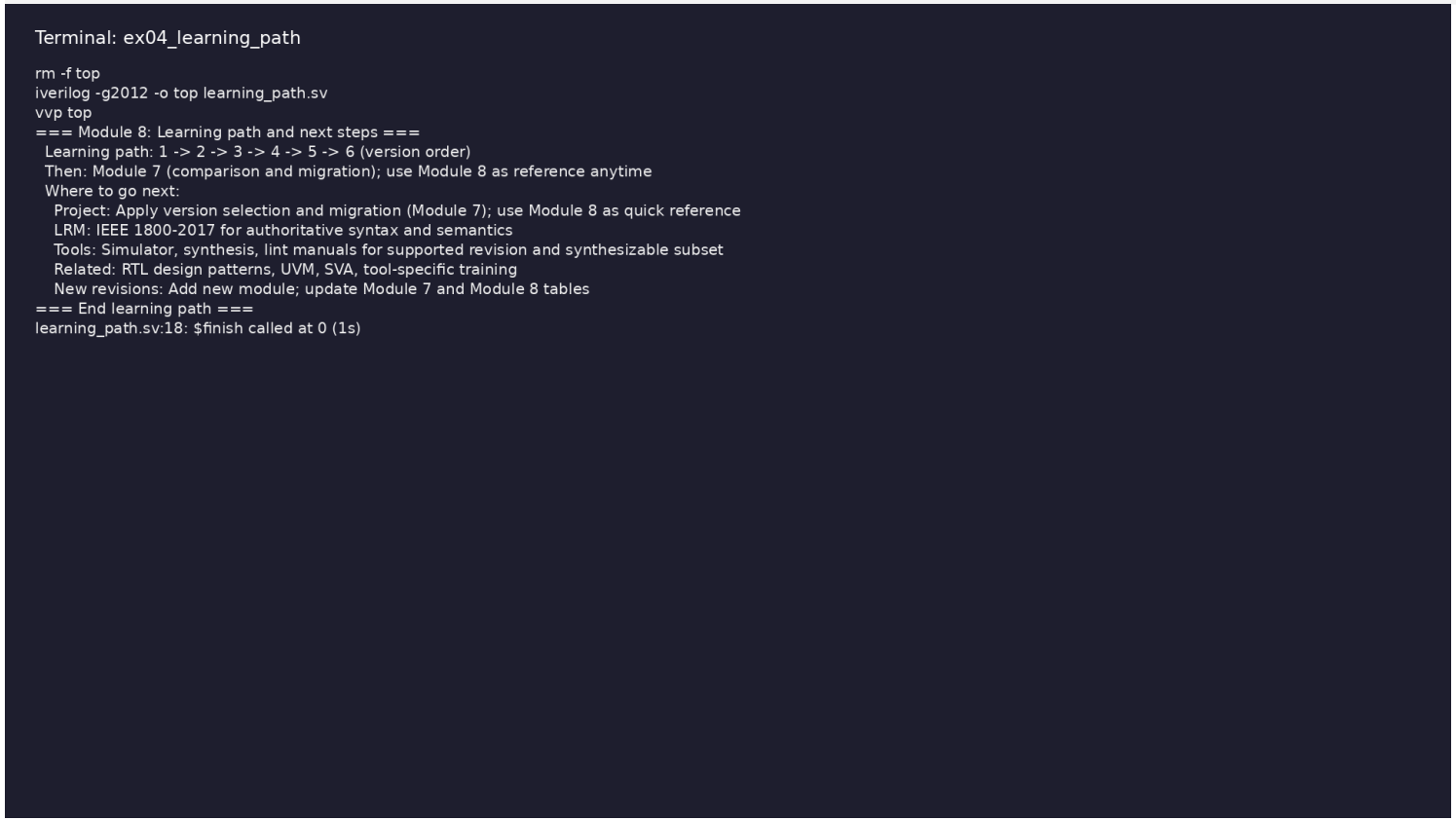
```
$ cd module8/examples/course_map && make clean && make run
```

# Demo: Design Subset

```
$ cd module8/examples/design_subset && make clean && make run
```

# Demo: Learning Path

```
$ cd module8/examples/learning_path && make clean && make run
```



# Demo: Migration Steps

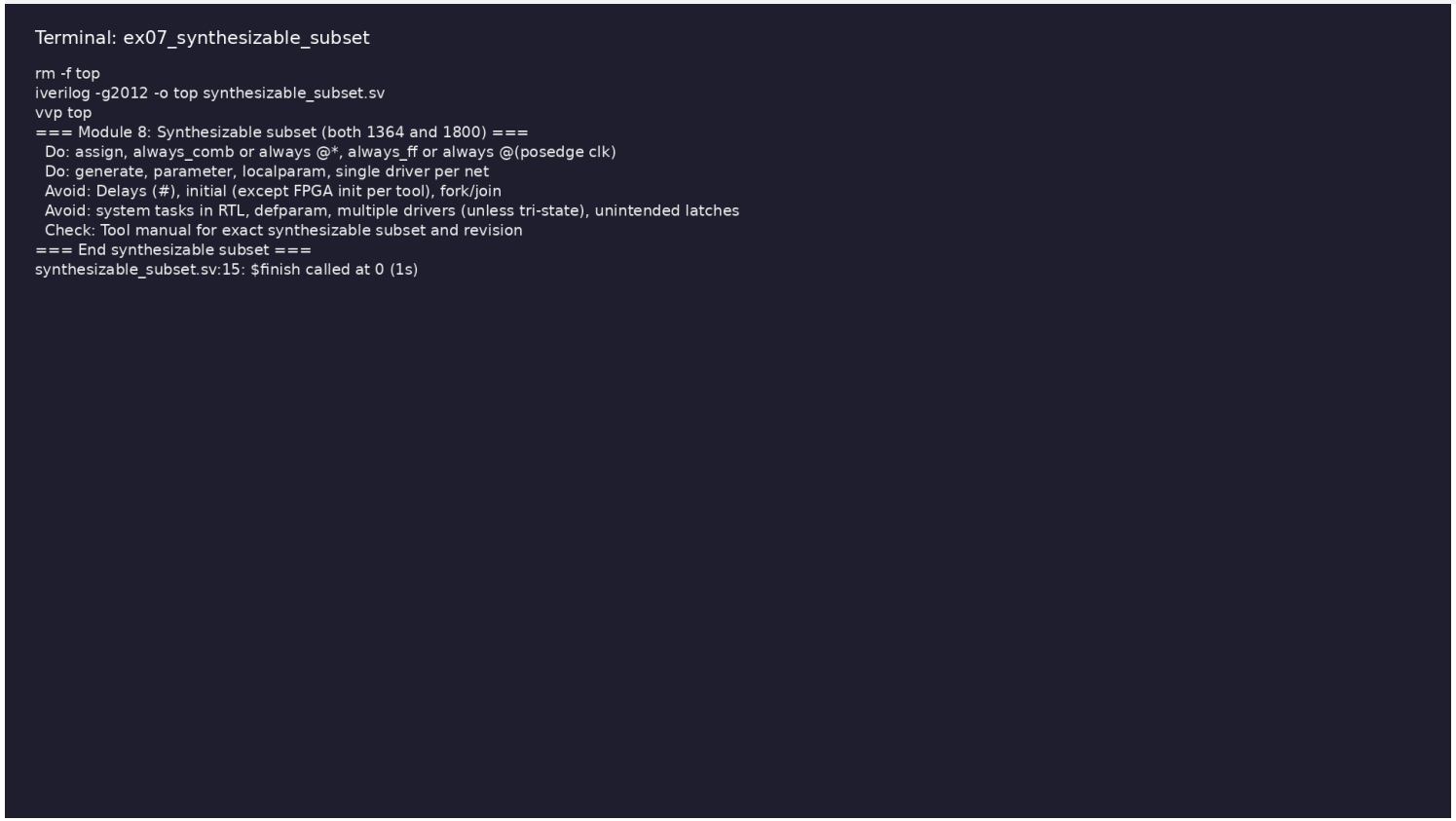
```
$ cd module8/examples/migration_steps && make clean && make run
```

# Demo: Pitfalls

```
$ cd module8/examples/pitfalls && make clean && make run
```

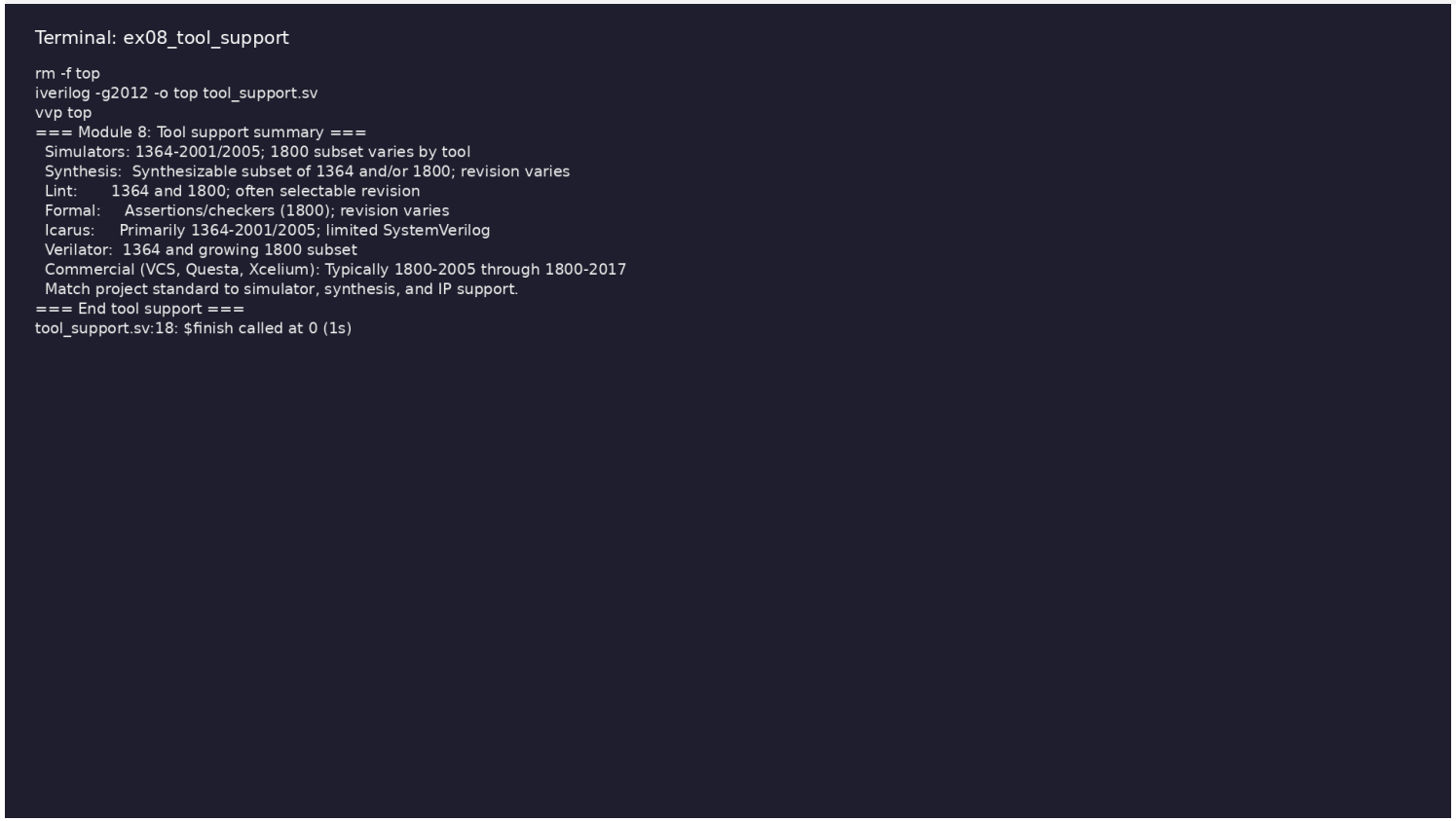
# Demo: Synthesizable Subset

```
$ cd module8/examples/synthesizable_subset && make clean && make run
```



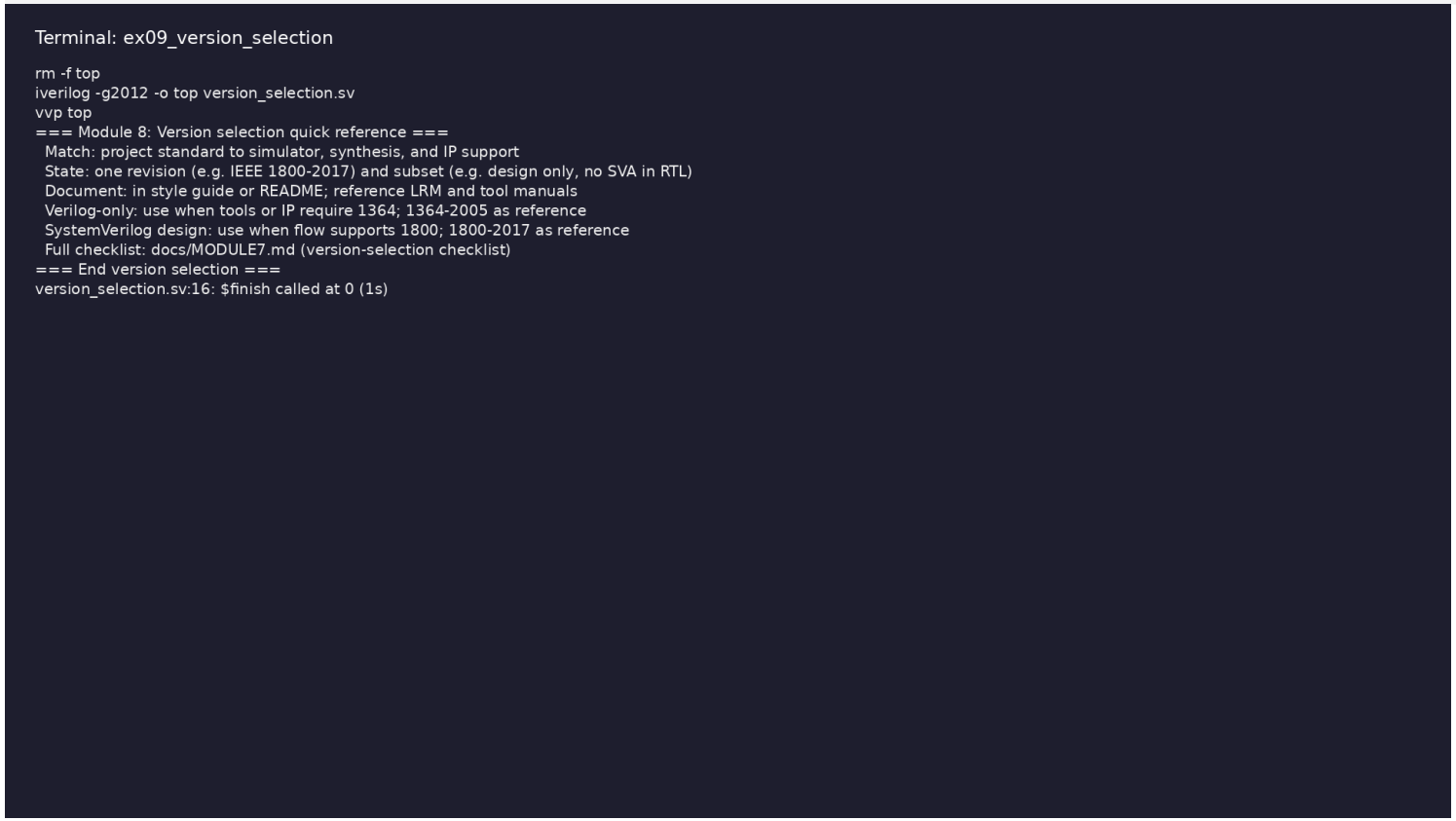
# Demo: Tool Support

```
$ cd module8/examples/tool_support && make clean && make run
```



# Demo: Version Selection

```
$ cd module8/examples/version_selection && make clean && make run
```



# Demo: Version Timeline

```
$ cd module8/examples/version_timeline && make clean && make run
```

# Practice & assessment

# Exercises

- Lookup
- Subset
- Migration
- Course map
- Cheat sheet

# Exercises

- Assuming a construct is in an older standard
- Mixing subsets without a rule
- Skipping regression after migration
- Forgetting tool support
- Treating this module as a replacement for Modules 1–7

# Summary & next steps

- Provide a single quick reference and course summary for Verilog and SystemVerilog across IEEE versi…
- Complete module8/CHECKLIST.md
- Run ./scripts/module8.sh --check
- Next: Next module in course